R13

Code No: 115AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, February - 2022
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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used paper feb-22 jntu

1.a) Compare and contrast the functional Programming and Logic Programming.
b) Explain the EBNF for common programming languages features. [8+7]

2.a) What is the difference between structured programming language and object-oriented programming language? Explain with examples.
b) List and explain the different programming domains and their associated languages. [8+7]

3.a) Define array. Explain various array operations with an example.
b) Explain about type compatibility. [10+5]

4.a) What are various conditional statements? Explain with an example.
b) Explain the Arithmetic relational and Boolean expressions. [8+7]

5.a) List and explain the design issues for functions user-defined overloaded operators.
b) Explain about local referencing environments. [10+5]

6.a) Explain the subprograms in a programming language.
b) Explain the static and dynamic scope. [10+5]

7.a) What is the importance of logic programming? Discuss the basic elements of Prolog. Give the applications of logic programming.
b) Explain about semaphores. [10+5]

8.a) Explain the use of conditional expression in Haskell's language.
b) Discuss the functional and imperative languages. [7+8]

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